

Risk management & financial regulation

Week 42 — what a euro in the bad state is worth, and to whom

Advanced Corporate Finance (EBM232A05)

University of Groningen · Sebastian Pfeil

Five rows. Every friction so far changed what a firm *does*:

Wk	Friction	Effect	Evidence
36	Personal taxes (Miller)	value $\uparrow / \downarrow$	net benefit (Korteweg, 2010)
37	Dynamics / flexibility	leverage $\downarrow$	flexibility survey (Graham and Harvey, 2001)
38	Costly external finance	investment $\downarrow$	cash hoarding (Almeida et al., 2004)
39	The manager wedge	risk-taking $\downarrow$	weak incentives (Jensen and Murphy, 1990)
41	Free-riding on control	takeovers $\downarrow$	acquirer gains ≈ 0 (Moeller et al., 2005)
42	Someone else's downside	? today	—

Last week the market could not discipline a firm by itself. When that happens, somebody writes **rules** instead — and to see why, we first have to ask what a bad outcome is actually *worth* to the person choosing.

Today

- The half you already own: under MM a costly hedge **destroys** value, and equity is a **call option** on the firm's assets.
- The puzzle: firms hedge **anyway** — three in five, for decades — and what hedging they do tracks **who the manager is**, not what the firm needs.
- One question, asked twice: **what is a euro in the bad state worth to whoever is deciding?**
- For a constrained firm the answer is **more** — and that is the entire case for hedging (Froot et al., 1993).
- For an insured bank the answer is **nothing** — and that is the entire case for regulation (Merton, 1977).

Same question, opposite answers. That difference is exactly why one firm hedges itself and the other has to be **made** to.

The half you already own

Hedging is worth nothing — and you can prove it

Modigliani–Miller does not stop at capital structure. Run the same argument on risk:

- A hedge is a **fairly priced** contract, so buying one is a zero-NPV transaction, like any other.
- Diversifiable risk is **not priced**, so removing it does not lower the cost of capital — and a shareholder who wanted less exposure could hedge **in her own account** anyway (*home-made hedging*, exactly like undoing leverage).
- And the contract may be fair, but the **apparatus is not free**: spreads, collateral, bank fees, a treasury desk.

Fair bet *minus* real costs is **negative** NPV — so under MM the hedging CFO is not wasting his time, he is destroying value, deliberately and every quarter. That is what makes the next slide a puzzle.

And you can already draw a bank

You also own the option machinery, and you have used it on exactly this shape:

- Equity in a levered firm is a **call** on the firm's assets, struck at the face value of the debt. You computed this.
- Option values **rise with volatility**. You computed that too, and called it **vega**.
- And in Week 39 you saw a board **buy** risk-appetite with convex pay — and overshoot.

None of that is re-taught today. Today we put a **second party underneath** the call — somebody who agreed to absorb the losses — and the same picture yields a conclusion it has never yielded before.

**The puzzle: hedging looks like a
hobby**

And yet they do it — almost everywhere

Bartram et al. (2009) read the accounts of 7,319 non-financial firms across 50 countries and simply counted who uses what:

Uses <i>any</i> financial derivative	$\approx 60\%$
... currency derivatives	$\approx 44\%$
... interest-rate derivatives	$\approx 33\%$
... commodity-price derivatives	$\approx 10\%$

Behind those numbers sit treasury desks, ISDA agreements and posted collateral — a permanent, staffed, budgeted activity.

Three firms in five run a policy the previous slide says **destroys value**, and have run it for decades. **That** is the puzzle — not that they hedge too *little*, which means nothing until some theory says hedging is worth anything at all.

And the pattern points at the manager, not the firm

Tufano (1996) picked the cleanest possible laboratory: North American gold miners, an industry with **one** dominant risk, fully observable hedging, and 48 firms doing visibly different things.

- He tested the **shareholder-value** theories — distress costs, tax convexity, investment policy (Smith and Stulz, 1985) — and found **little support for any of them**.
- What did predict hedging: **managers holding stock hedge more; managers holding options hedge less**. Risk management also falls with CFO tenure.

This is the Week-39 wedge again, in a new place: the policy tracks the **decision-maker's own exposure**, not the firm's. And you already have the causal version — Bakke et al. (2016) cut option pay exogenously and hedging **rose**.

So we need a reason that is not a hobby

We are in an uncomfortable position, and it is worth naming:

- MM says a costly hedging programme **destroys** value — and three firms in five run one anyway.
- The one serious attempt to explain that with **shareholder** value failed, and what survived was **who the manager is**.
- So: is there *any* reason a **shareholder** should want the firm to hedge?

What is a euro in the bad state worth to whoever is deciding?

That is today's only question. Ask it of a cash-strapped firm and you get the case for hedging. Ask it of an insured bank and you get the case for regulation. The answers point in **opposite** directions.

A euro in the bad state is worth more

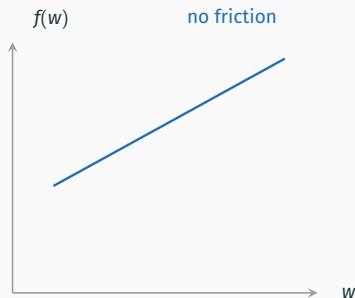
The setup, with a friction you already own

One firm, one period, two states. Internal cash w arrives high or low, equally likely. There is an investment opportunity costing I and adding $G > 0$.

- If $w \geq I$, the firm funds it out of pocket.
- If $w < I$, it must raise the shortfall $I - w$ outside, at κ per euro — the **same** κ you met in Problem Set 37 and again in Week 38. Nothing new is assumed here.
- So firm value is $f(w) = w + G - \kappa(I - w)^+$.

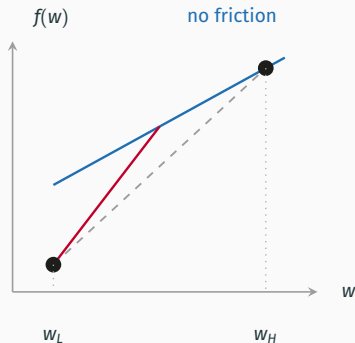
Read that expression once more. Below I , an extra euro of **own** money is worth more than a euro — it also saves you κ of financing cost. Above I , it is worth exactly a euro. That is the whole model.

What hedging is worth



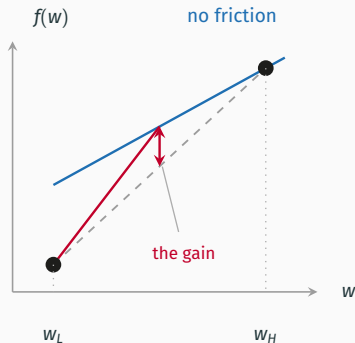
Start with MM. Set $\kappa = 0$ and value is **linear** in internal funds: $f(w) = w + G$. A straight line is precisely the statement that spread does not matter — the average of the two ends *is* the value at the middle. So even a **free** hedge would change nothing here, and real ones are not free.

What hedging is worth



Now switch the friction on. Below the line **bends down**: every euro short has to be bought at κ . The firm's two states sit at w_L and w_H , and what it expects without hedging is the **dashed chord** between them — which, under a concave function, lies **below** the curve.

What hedging is worth



Hedging replaces the two states by their **average** — one point, on the curve. The gap between the curve and the chord is the gain, and it is nothing but the **concavity**. No risk premium was earned; a deadweight financing cost was avoided.

Hedging pays because value is concave in cash

Costly outside money makes f **concave**, so for a mean-preserving contraction

$$f(\mathbb{E}[w]) > \mathbb{E}[f(w)] \implies \text{hedge.}$$

- Nothing here is about **risk aversion**. The shareholder is risk-neutral; the concavity is a **technology**, not a taste.
- The hedge is still **fairly priced** — so slide 4 stands. What changed is **which states the friction bites in** (Froot et al., 1993).

In Week 39 the payoff was **convex**, the chord sat *above* the curve, and the manager therefore **wanted** variance. Same picture, curvature flipped, conclusion reversed.

Put in numbers

Let $I = €100$, $G = €40$, $\kappa = 0.5$; cash is €40 or €160, equally likely.

	Bad state	Good state
Internal cash w	40	160
Financing cost, $\kappa(I - w)^+$	30	0
Firm value $f(w)$	50	200
No hedge: $\mathbb{E}[f(w)]$		125
Hedge to $\mathbb{E}[w] = 100$: $f(100)$		140
Gain from hedging		15

The €15 is exactly half the €30 the bad state would have paid in financing cost — the deadweight loss, avoided with probability one half. Hedging did not create value; it **stopped** some from being destroyed.

The result nobody guesses: it depends on the correlation

So far the opportunity was the same size in both states. Drop that, and let the investment opportunity G be **good** exactly when cash is **high** — correlation $\rho > 0$.

- Then the firm's money already arrives **when it has something to spend it on**. The mismatch hedging was correcting is **smaller**.
- If $\rho < 0$ — opportunities arrive precisely when cash does not — the mismatch is **worse**, and the firm should hedge **more** than fully.
- Full hedging is optimal only in the knife-edge case $\rho = 0$.

So “how volatile is my cash flow?” is the **wrong question**; the right one is “**does my cash arrive when I need it?**” And the second fact from the accounts now reads differently: firms cover only a **fraction** of disclosed exposure, which under this model is not a failure to hedge — it is $\rho > 0$.

Which is why gold miners hedge so little

Return to Tufano (1996)'s puzzle with the model in hand. What is a gold miner's investment opportunity worth when the gold price is **high**?

Gold price	Cash flow	Value of developing a new seam
High	high	high
Low	low	low

Cash and opportunities are **strongly positively correlated** — the $\rho > 0$ case — so the model's prescription for such a firm is to hedge **less than fully**, which is what they do. Careful, though: that is the **model's** reading of Tufano's fact, not his own conclusion. It does mean low hedging in gold mining is **not** the clean refutation it looks like.

**The same euro, in someone else's
pocket**

Now change who receives the bad state

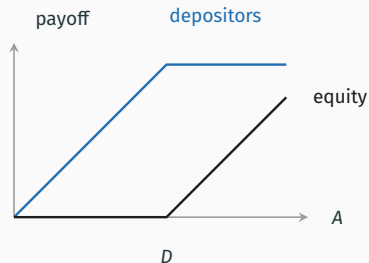
Everything so far assumed the decision-maker **lives in** the bad state. Take that away.

- A bank funds assets A with insured deposits D and capital $w = A - D$ — Week 38's w , deliberately: what the chooser eats first.
- Depositors are made whole **whatever happens**: they do not monitor, do not charge for risk, do not care what the bank buys.
- Shareholders keep what is left, and **limited liability** covers them when there is nothing.

What is a euro in the bad state worth to whoever is deciding?

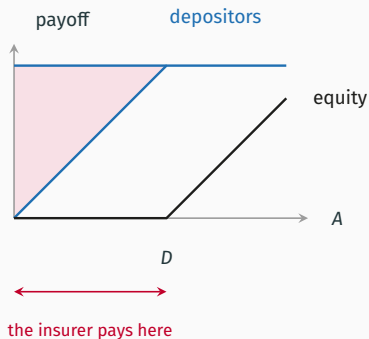
Same question as ten minutes ago — the answer comes back **opposite**. (Week 36 was **Merton Miller**; today is **Robert Merton**. Both 1977.)

Who gets what



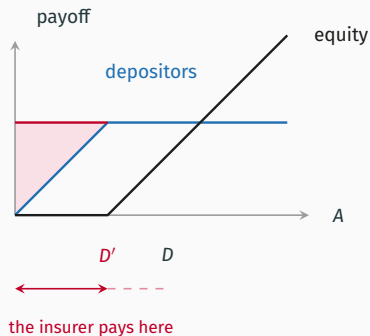
With **no** insurance, the split is the one you already know. Depositors receive $\min(A, D)$ and eat every shortfall; equity receives $\max(A - D, 0)$ — a **call struck at D** . Because depositors bear the downside, they **price** it.

Who gets what



Insurance arrives, and depositors are flat all the way across: they get D in **every** state. Somebody pays for that — the shaded region is $\max(D - A, 0)$, a **put the insurer wrote**. Now look at what did **not** move: equity's line is untouched.

Who gets what



Raise capital and the strike slides **left**. Both lines move: the triangle has area $D^2/2$, so exposure falls with the **square** of the strike — and equity is now paid in states where it used to get **nothing**. Capital is the **deductible**.

The safety net creates the risk it insures

Deposit insurance is a written put of value S , and

$$\frac{\partial S}{\partial \sigma} > 0 \quad \text{while} \quad \frac{\partial(\text{equity})}{\partial \sigma} > 0 \text{ by the same amount.}$$

- Equity is a call, so it gains from volatility — the **vega** you already own (Merton, 1977).
- The gain is not created. It is taken, euro for euro, from the party that wrote the put.
- And **nobody objects**: the depositors who would normally push back have been insured out of caring.

A euro in the bad state is worth **nothing** to this decision-maker. So he will pay to make the bad state **worse**.

Put in numbers

Assets worth €100 in expectation, deposits $D = €92$, so capital is €8 — an 8% capital ratio. Compare two asset portfolios with the **same** expected value.

	Safer: {70, 130}	Riskier: {50, 150}
Expected assets	100	100
Expected value of equity	19	29
Expected cost to the insurer S	11	21
Change	+10 to equity	−10 to the insurer

The bank produced **nothing** — expected assets are €100 either way. It simply reached across and took €10 from the taxpayer, and any banker who declined to do it would be underperforming his peers.

Capital is the deductible

Repeat that experiment at a different capital ratio. Same assets, same risk shift, only w changes:

Capital w	Deposits D	Gain to equity from the shift	Cost to the insurer
8	92	+10	10
20	80	+10	10
40	60	+5	5

Look at the middle row: **tripling capital from 8 to 20 changes the incentive not at all**, because equity is wiped out either way. Nothing moves until capital passes **30** — the bad state of the *safer* portfolio. (A different question from the figure, and a different shape: there we asked what the insurer expects to *pay*, which falls smoothly; here, when the **incentive** moves, which is a step.) **Thin buffers buy nothing.**

The model in public

Silicon Valley Bank, March 2023

A bank with roughly \$209bn of assets, and a depositor base unlike anyone else's:

	SVB	Peer average
Uninsured deposits, % of total (end-2022)	94%	41%

- Uninsured depositors are the ones the model says **do** still care — they are not covered, so they monitor, and they run.
- On **9 March** over \$40bn left in a single day; management expected over \$100bn more the next morning. The bank was closed on **10 March 2023**.

Note which way round this is: SVB failed **because** so much of its funding sat *outside* the safety net. Insurance suppresses runs — that is what it is for. (Board of Governors of the Federal Reserve System, 2023, Figure 7 and pp. 1–2)

And then the put was rewritten over a weekend

Under the ordinary rule, those uninsured depositors take a loss above \$250,000. That is not what happened.

- The authorities invoked the **systemic-risk exception** and guaranteed **every** depositor, insured or not.
- So a put written on 6% of the deposit base was, in a weekend, converted into a put on **100%** of it — at a cost to the insurance fund of roughly \$16bn (Office of Inspector General, Board of Governors of the Federal Reserve System, 2023).
- Every bank and every large depositor in the system watched that happen.

This is the comparative static performed in public. The model says a wider put means weaker monitoring and more risk-taking next time — and the decision to widen it was taken, reasonably, to stop a panic. **Both things are true.**

What the model asks of a regulator

For sixty years, the premium ignored the risk

If the insurer is writing a put, it should charge for it — and the price should depend on σ and on w , because those are what the put is worth. It did not.

1933–1992	Every insured bank pays the same flat rate
From 1 Jan 1993	Nine risk classes , priced on capital and supervisory rating

Federal Deposit Insurance Corporation (2024). A flat premium is not merely too low for the risky bank. It is simultaneously **too high** for the safe one — so the safe, well-capitalised bank is **paying part of** its riskier competitor's insurance.

A cross-subsidy running from prudent banks to imprudent ones, for sixty years, by construction. The 1993 reform is this model, legislated.

Two instruments, and why you need both

Price the put (risk-based premiums):

- Charges σ directly.
- But the regulator must **measure** σ , and the bank knows it better.

Raise the deductible (capital):

- Works **without** measuring σ — equity simply bears the first loss.
- But only once the buffer survives the bad state.

Capital requirements are the **Basel** framework's answer, and the reason the debate is always about the **level**: too thin and it is not a deductible at all.

Both instruments are aimed at one target — putting the bad state back into the pocket of whoever chooses σ . That is the only thing this week has been about.

Ledger and wrap

Risk management is not about disliking risk. It is about **who receives the bad state**. Give the bad state to a firm that needs its own cash to invest and it will pay to smooth it. Give it to somebody else and the firm will pay to make it worse.

Two things to carry forward:

- “Identify the friction” now includes **who bears the downside** — and that party is not always inside the firm.
- Every fix works by **restoring the loss to the chooser**: hedging does it voluntarily, capital does it by force. An instrument that fails to do it — a thin buffer, a flat premium — changes nothing at all.

The frictions ledger — Week 42

Six rows. One more to come.

Wk	Friction	Effect	Evidence
36	Personal taxes (Miller)	value $\uparrow$ / $\downarrow$	net benefit (Korteweg, 2010)
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39	The manager wedge	risk-taking $\downarrow$	weak incentives (Jensen and Murphy, 1990)
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42	Someone else's downside	hedging $\downarrow$ / risk $\uparrow$	gold miners (Tufano, 1996)

This is the first row where the friction points in **two directions at once** — too little risk management in the firm, too much risk-taking in the bank. Same friction, and the sign depends only on **whose** downside it is.

What is new here — and what is not

You already own:

- MM, extended to hedging
- equity as a call; vega
- that convex pay buys risk appetite

This week adds:

- value **concave in cash** — and the correlation, not the variance, decides
- the **written put**, and who pays for it
- capital as a **deductible**, with a floor below which it does nothing

The machinery was all yours already. What is new is that it was pointed at a party **outside** the firm — and the answer came back as a policy question.

Reading, and next

- **Reading:** Froot et al. (1993) — the introduction and Sections I–II carry the argument; today's two-state version is the scaffold for the rest.
- **For the banking half:** Merton (1977) is nine pages and mostly algebra you do not need — read it for the **claim**, which is the one on our figure.
- **Evidence:** Tufano (1996), and read his Table of motives **like a referee**: which theories could that sample have rejected, and which could it never have tested?
- **This week:** the **replication project is due**. Bring the repository that runs, not the one you meant to write.
- **Next:** Week 43, payout. A firm that can cut its dividend and does not — and what that refusal is buying.

Nobody hedges a risk that somebody else has agreed to carry.

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